

Problem 2: “WORM” Search

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Setup

You successfully helped the Nematode Observation Division debug the WIGGLE system, but before you get a chance to settle into your new lumbar-supported office chair you feel a tap on your shoulder. The chief of the trematode monitoring program sheepishly asks, since you did so well with the WIGGLE system, if you’d help try making sense of outputs of their new Global Unit for Trematode Surveillance (GUTS).

The technicians on duty explain that the unit’s patented universal pattern detector has become fascinated with a peculiar word search generated from tematode genomic signal noise. It insists that the key pattern hidden within the data is the word **WORM**. We want to take it’s large output of readings, and find the number of times this word appears in them.

This particular word search is flexible: words may appear horizontally, vertically, or diagonally, written backwards, or even overlapping other words. The program’s challenge is to count *all* occurrences of WORM in every direction.

Here are a few illustrative examples of how WORM might appear (non-relevant letters replaced with X’s):

XXWXXX
XRMOWX
XOXXRX
WORMXM
XWXXXX